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PALEOMAGNETISM OF THE LATE CRETACEOUS-EARLY TERTIARY NORTH-CENTRAL MONTANA ALKALIC PROVINCE

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Abstract. Paleomagnetic results have been obtained from the Late Cretaceous-early Tertiary igneous complexes of the north-central Montana alkalic province. Data from 94 sites in Eocene volcanic and intrusive rocks give a paleomagnetic pole located at 82.0°N, 170.2°E (A_{95} = 3.5°; k = 18.6), while 36 sites in Paleocene intrusions yield a paleomagnetic pole at 81.8°N, 181.4°E (A_{95} = 5.4°; k = 20.2). These poles differ by only 1.6° and are not significantly different statistically. The 130-site virtual geomagnetic poles show no significant elongation and suggest no significant apparent polar wander (APW) during the period of magnetization of the igneous centers. Postmagnetization structural complications in these rocks are minimal. The presence of a single predominant polarity in these intrusive complexes reinforces the radiometric age data that suggest that igneous activity within individual centers was of short duration. The north-central Montana data together with other early Tertiary, Cretaceous, and mid-Tertiary paleomagnetic results require modification of our earlier APW chronology [Diehl et al., 1980]. It now appears that APW relative to North America since the Early Cretaceous consists of a polar stillstand during much of the Cretaceous (120-75 m.y. B.P.), a period of rapid movement in latest Cretaceous time (75-65 m.y. B.P.), and a period of slow polar movement thereafter. The onset of this period of rapid APW correlates well with a major change in plate motions at ∿80 m.y. and the beginning of the Laramide Orogeny.

Introduction

Paleomagnetic pole determinations initially helped to delineate large-scale relative motion between stable cratons; the discovery of these movements was vital to the formulation of the theory of plate tectonics. During the last decade, much interest has centered on using paleomagnetism to study the detailed interaction of major plates at their boundaries. It has been shown, for instance, that interaction between major plates may involve accretion, rotation, and possibly large-scale translations of oceanic and continental fragments [Beck, 1976, 1980; Irving, 1979; Jones et al., 1982]. Paleomagnetism provides a most useful method for studying the tectonic histories of these crustal fragments, displaced terranes, or "microplates" because relatively minor rotations can be detected by deflected remanent declinations and northward or southward translation is recorded by anomalous inclinations. However, the ability of paleomagnetic studies to elucidate the tectonic histories of these fragmented blocks requires a set of reference poles that accurately determine the apparent polar wander (APW) path for the time interval of concern.

Ages of many of the allochthonous blocks from the North American Cordillera fall in the interval between the Late Cretaceous and Oligocene. The Cretaceous paleomagnetic pole for the North American craton is well established from studies of rocks older than 75 m.y. [Mankinen, 1978] and is located at 68.0°N, 186.0°E (A_{95} = 2.2°). Beck et al. [1977] in reviewing mid-Tertiary poles for North America calculated a mean Oligocene paleomagnetic pole at 81°N, 132.5° E (A_{95} = 4.0°). Eighteen degrees of arc separate these two paleomagnetic poles. Studies as recent as 1977, of Eocene rocks [Shive and Pruss, 1977; Lovlie and Opdyke, 1974; Shive et al., 1977] yield pole positions that are very near the Oligocene paleomagnetic pole. Therefore, during this crucial interval (circa 75-45 m.y. B.P.) in the early Tertiary, a large amount of APW must have occurred relative to North America. Although McElhinny [1973] lists seven early Tertiary pole positions, APW during this time interval is imprecisely known because very few of these poles were based on enough sampling sites to be considered reliable. As pointed out by Butler and Taylor [1978], this interval of large apparent pole migration is also coincident with the Laramide orogeny, and therefore detailed knowledge of the behavior of the apparent pole during the early Tertiary is essential in order to determine the degree to which this orogenic event is reflected in the APW path. Irving and Park [1972] have already shown that "hairpins" in the North American apparent polar wander path correlate with orogenic events resulting from plate interactions.

With these problems in mind, we began a comprehensive study of the early Tertiary igenous rocks of the north-central Montana alkalic province. Diehl et al. [1980] have previously reported preliminary results for two early Eocene intrusive centers: the Highwood and Bearpaw mountains. Results from the Paleocene intrusions of the province were reported by Jacobson et al. [1980]. Utilizing the Montana data and recent paleomagnetic results from Arizona [Barnes and Butler, 1980] and Wyoming [Sheriff and Shive, 1980; Shive and Pruss, 1977], Diehl et al. [1980] suggested that APW relative to North America since Early Cretaceous time consisted of three periods of stillstand (little apparent polar motion) separated by episodes of relatively rapid polar movement. The periods of rapid polar

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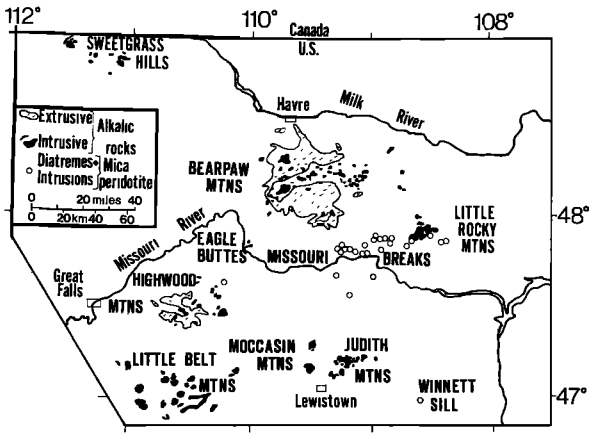


Fig. 1. Index map, modified from Marvin et al., [1980].

movement appeared to have occurred between roughly 75-65 m.y. B.P. and again perhaps 44-35 m.y. B.P. and were thought to correlate with major changes in plate geometries and velocities that occurred at about 80 and 40 m.y. B.P., as suggested by Coney [1972, 1978]. In this paper we analyze results for all intrusive centers of the north-central Montana alkalic province, including those previously reported. This requires that the timing of APW by Diehl et al. [1980] be modified.

Geologic Setting

An early Tertiary age for the igneous rocks of the north-central Montana alkalic province is well established. Extensive K-Ar, fission track, and U-T-Pb dates [Marvin et al., 1980, 1973] indicate that peak igneous activity occurred between 55 and 50 m.y. ago with major intrusions also occurring during the Late Cretaceous and Paleocene, 69-60 m.y. ago. The major igneous centers are shown in Figure 1. Ages fron individual centers suggest that activity in each area lasted approximately 3-7 m.y. and that no systematic geographic migration of this igneous activity took place.

A brief description of the geology of each igneous center is given below. All ages cited are from Marvin et al. [1980] with the exception of those from the Little Belt Mountains which are from Marvin et al. [1973], which are recalculated with new International Union of Geological Sciences decay constants.

The Sweetgrass Hills (54-50 m.y. B.P.) are the most northerly of the intrusive/volcanic centers that make up the north-central Montana alkalic province. The igneous rocks are entirely intrusive and are primarily concentrated in three prominent topographic highs: West, Middle, and East buttes. A few scattered small intrusions also occur in the surrounding plains. The Sweetgrass Hills have been mapped by Kemp and Billingsley [1921]. Truscott [1977] worked out the intrusive sequence of East Butte where the igneous activity appears to have lasted at least 2 m.y. [Marvin et al., 1980]. Rock types present in the Sweetgrass Hills are syenites, trachyte, vogesite, minettes, rhyolite, aplite, and monzonite: The more mafic rocks are located in the western portion, while the more alkali-rich rocks occur in the eastern portion.

The Bearpaw Mountains (54-50 m.y.) consist of a northern and southern volcanic field separated by the elongate Bearpaw Mountains arch [Hearn, 1976]. The arch is a composite anticlinal uplift containing numerous stocks, laccoliths, sills, and dikes. These intrusions are composed of biotite pyroxenite, syenite, shonkinite, nepheline syenite, monzonite, porphyritic latite, analcime phonolite, tinquaite, and carbonatite. The volcanic fields consist of flows and pyroclastic rocks of interlayered mafic phonolite and porphyritic latite which are unconformably overlain by analcime trachyte and mafic analcime phonolite flows of lesser volume. Intrusions also occur within the volcanic fields. Extensive gravity sliding [Reeves, 1946] and related collapse, associated with the uplift of the central arch, have modified the original structure so that the volcanics now dip toward the arch at an average angle of 35°.

The Highwood Mountains (53-49 m.y. B.P.) are similar to the Bearpaws in that they also consist of both intrusive and extrusive rocks, have a felsic and mafic series, and have been affected in part by gravity sliding. The earliest volcanic activity produced quartz latite flows, breccias, conglomerates, and tuffs deposited over a terrain of considerable relief [Larson, 1941]. This volcanic pile was then eroded to a hilly terrain before being covered by mafic phonolite flows [Larson, 1941]. With the exception of the amphibole phonolite dikes which appear to be related to the earliest volcanic episode [Buie, 1941], the intrusive rocks of the Highwoods are contemporaneous with the second stage of volcanic activity which produced the mafic phonolite flows [Hurlbut and Griggs, 1939; Burgess, 1941]. Stocks, sills, dikes, and laccoliths were intruded into flat-lying Cretaceous sandstones and shales and into the associated volcanics [Buie, 1941; Burgess, 1941; Hurlbut and Griggs, 1939]. These intrusions consist of amphibole phonolite, shonkinite, syenite, nepheline syenite, biotite pyroxenite, and mafic phonolite [Woods, 1974; Burgess, 1941; Hurlbut and Griggs, 1939; Buie, 1941].

The igneous rocks of the Little Belt Mountains occur primarily in laccoliths in the northern and central portions of the area, while thick sills predominate in the southern part. Other shallow-level intrusions such as stocks, dikes, and plugs also are present. These intrusions range in composition from rhyolite and granite porphyry through intermediate rocks, mainly quartz latite porphyry, to mafic rocks such as shonkinite and lamprophyre. The Little Belt rocks have been studied in detail by Witkind [1971, 1973] and Keefer [1972]. Radiometric dates indicate that intrusive activity began in the southern portion of the Little Belts and that the intensity of igneous activity increased as it progressed northward [Marvin et al., 1973]. With the exception of a small vogesite dike that yields a hornblende Paleocene date that is possibly too old, most of the intrusions were emplaced between 49 and 55 m.y. ago with some magmatic activity continuing to 42 m.y. ago.

The Eagle Buttes igneous complex consists of

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numerous small dikes, sills, and plugs that are densely clustered about four main laccolithic intrusions. Both Lindvall [1962] and Gucwa [1971] have mapped the area. The complex lies midway between the Bearpaw Mountains and the Highwood Mountains and contains rocks similar to those of the Bearpaws: porphyritic shonkinite, syenite, and porphyritic hornblende latite. Radiometric dates from the complex range from 51.7 to 49.6 m.y.

The dikes, plugs, and breccias of alkalic ultramafic rock (monticellite peridotite and alnoite) and kimberlite associated with the Missouri Breaks diatremes have been mapped by Hearn [1968, 1979]. These igneous features form a broad zone approximately 140 km long between the Highwood and the Little Rocky mountains. Other isolated occurrences of similar rocks (the Winnett Sill among others) occur southeast of the Little Rockies [Ross, 1926; Johnson and Smith, 1964]. Emplacement of these rocks occurred between 52 and 47 m.y. ago. An isolated outbreak at Smokey Butte 27 m.y. ago farther east marks the waning phase of igneous activity in the north-central Montana alkalic province.

The Moccasin Mountains consist of North and South Moccasin Mountain, each a laccolith with associated dikes and sills [Blixt, 1933; Miller, 1959; Lindsey, 1982]. North Moccasin Mountain is composed of syenite porphyry, alkali syenite porphyry, and quartz monzonite porphyry. Intrusive activity appears to have occurred between 66 and 64 m.y. with hydrothermal activity and mineralization continuing to 53.5 m.y. ago.

The oldest rocks of the north-central Montana alkalic province are located in the Judith Mountains. Radiometric ages from this intrusive complex indicate that igneous activity occurred sporadically from late Cretaceous (69-68 m.y.) to Eocene (47 m.y.), an interval of time much longer than any other igneous center of the alkalic province. The geology of the Judith Mountains has been mapped by Weed and Pirsson [1898] and Wallace [1953, 1956]. Forrest [1971], Hall [1976], and Kirchner [1982] have mapped parts of the area. Quartz diorite, diorite, monzonite, quartz monzonite, rhyolite, syenite, alkali syenite, tinguaite, and alkali granite occur in the stocks, laccoliths, sills, and dikes that make up this intrusive complex.

The Little Rocky Mountains, like the Judith and Moccasin mountains, are the exposed core of a Paleocene intrusive center. Rock types are quartz syenite, syenite, alaskite, hornblende syenite, aegirine trachyte, and trachyte--all porphyritic. The geology of the Little Rocky Mountains has been described by Brockunier [1936], Knechtel [1959], and Bailey [1974]. The laccoliths or stocks, dikes, and sills of the Little Rocky Mountains appear to have been emplaced within the period of 67-60 m.y. ago.

TABLE 1. Distribution of Paleomagnetic Sites (Samples) for the North-Central Montana Alkalic Province

Igneous Complex	No. of Sites	No. of Samples
Little Rocky Mountains	24	212
Moccasin-Judith Mountains	17	152
Sweetgrass Hills	14	109
Bearpaw Mountains	46	441
Highwood Mountains	40	312
Eagle Buttes	8	66
Little Belt Mountains	14	121
Missouri Breaks diatremes	10	100

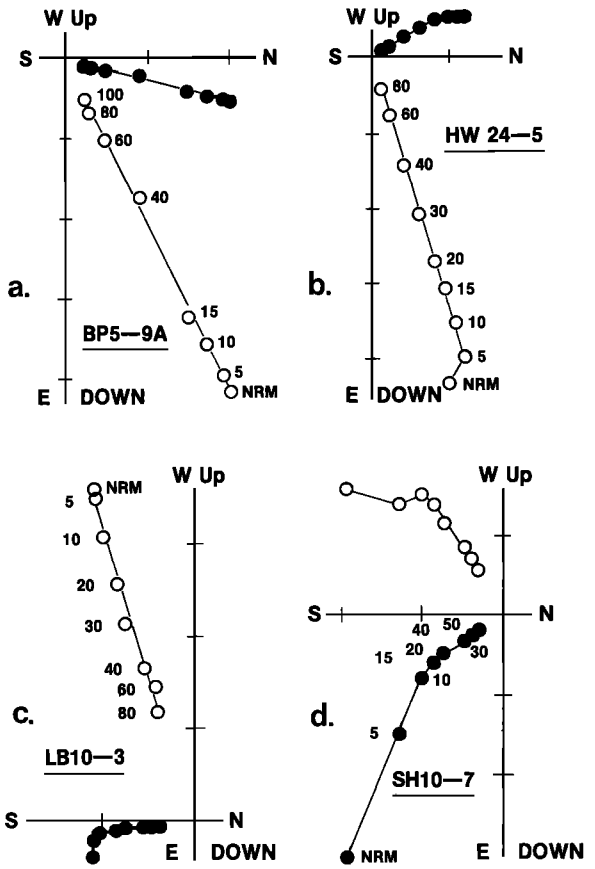


Fig. 2. Vector end point diagrams from representative samples of the north-central Montana alkalic province. Solid circles are projections on the horizontal plane; open circles are on the north-south vertical plane. The af demagnetization levels given in milliteslas. Horizontal and vertical axes divisions are (a) 5.0 x 10^{-1} A/m; (b) 2.0 x 10^{-1} A/m; (c) 5.0 x 10^{-2} A/m; (d) 1.0 x 10^{-2} A/m.

Paleomagnetic Data

A total of 1513 samples were collected from 173 paleomagnetic sites distributed geographically and lithologically throughout the nine major igneous centers that make up the north-central Montana province (Figure 1). Table 1 lists the number of samples and sites for each igneous complex. (For this study, the neighboring Moccasin and Judith mountains are treated as one igneous center.) Core samples were drilled with portable field equipment and oriented "in situ" with a magnetic compass and clinometer. Back azimuths and/or sun compass readings were taken

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TABLE 2. Summary of Paleomagnetic Data for North-Central Montana

Site	Lithology	λ	ϕ	N/No	Ho	D	I	α_{95}	δ	k	PLAT	PLONG	Age, m.y.
				Moccasin-Judith Mountains									
MJ1	syenite	47.18	109.32	9/9	200	169.1	-72.9	7.8	12.9	35.3	77.0	224.6	63.7
MJ2*	syenite	47.29	109.48	9/9	500	344.8	62.5	27.1	45.4	3.0	--	--	66
MJ3	syenite	47.30	109.48	9/9	500	337.2	62.0	14.5	24.2	10.2	73.5	154.3	66
MJ4	alkali granite	47.22	109.22	8/9	200	30.6	82.7	11.9	17.0	19.7	58.9	264.9	62.0
MJ5	quartz monzonite	47.20	109.22	8/8	400	219.5	-76.8	7.1	10.9	48.5	62.4	286.5	65
MJ6	alkali syenite	47.20	109.22	9/9	100	179.5	-58.2	5.2	8.6	79.4	81.7	73.5	67
MJ7	quartz monzonite	47.20	109.22	8/9	200	319.8	69.5	7.7	11.9	41.0	63.9	189.2	65
MJ8	alkali granite	47.22	109.22	8/9	200	353.6	71.4	2.2	3.4	508.0	80.3	228.8	62.0
MJ9	alkali granite	47.22	109.22	9/9	400	350.1	65.6	6.8	11.2	46.9	83.3	168.6	62.0
MJ10	trachyte	47.22	109.20	9/9	400	332.2	67.6	2.3	3.8	408.9	71.6	181.5	
MJ11	quartz monzonite	47.18	109.22	9/9	400	6.9	70.7	3.4	5.5	192.9	81.0	277.0	65
MJ12*	tinguaite	47.19	109.17	9/9	200	61.4	24.2	42.9	75.5	1.2	--	--	65
MJ13	alkali syenite	47.20	109.15	10/11	600	113.5	-76.1	9.2	16.0	23.2	51.1	210.4	67
MJ14	alkali syenite	47.22	109.15	10/10	400	166.9	-59.0	2.4	4.2	342.4	78.0	127.5	67
MJ15	rhyolite	47.18	109.23	10/10	400	333.6	80.7	2.0	3.6	465.5	62.4	233.3	
MJ16	quartz monzonite	47.17	109.22	9/9	300	349.7	63.4	3.9	6.4	142.9	82.5	147.0	65
MJ17*	quartz monzonite	47.18	109.31	/5									64.2
				Little Rocky Mountains									
LR1*	syenite	47.92	108.53	8/8	400	6.1	24.7	5.0	7.6	99.8	54.7	61.1	62.5
LR2	syenite	47.91	108.54	9/9	400	340.0	49.5	5.4	8.9	74.0	66.7	119.8	
LR3	syenite	47.96	108.55	5/8	300	332.4	52.1	11.7	13.6	28.6	64.3	135.3	
LR4	syenite	47.95	108.55	9/9	400	330.8	62.8	5.4	8.9	73.8	69.5	161.8	
LR5*	syenite	47.95	108.55	11/11	500	282.5	69.7	4.1	7.6	103.9	43.1	198.7	
LR6	syenite	47.94	108.55	9/9	400	347.8	58.2	4.8	7.9	92.9	77.4	120.3	
LR7	syenite	47.93	108.54	9/9	300	357.1	69.1	3.2	5.3	208.1	84.1	234.7	67.2
LR8	alaskite	47.93	108.54	9/9	200	343.7	61.1	8.1	13.3	33.0	77.2	140.7	
LR9	syenite	47.92	108.62	8/8	400	7.4	69.9	5.2	7.9	91.3	82.5	286.9	62.4
LR10	syenite	47.92	108.57	8/8	400	0.9	46.5	6.2	9.4	64.5	69.9	69.0	
LR11	syenite	47.92	108.60	8/8	400	22.4	73.1	5.5	8.4	81.5	73.0	293.8	
LR12	trachyte	47.90	108.62	7/8	200	12.7	68.2	6.1	8.7	74.9	81.1	314.3	61.0
LR13	syenite	47.95	108.64	7/7	400	316.1	52.5	4.9	6.9	118.9	54.1	153.4	
LR14	syenite	47.96	108.55	9/9	400	323.9	52.7	5.5	9.0	71.8	59.3	146.2	
LR15*	syenite	47.94	108.56	10/10	500	261.5	46.9	29.9	53.9	2.2	--	--	
LR16	syenite	47.95	108.57	11/11	400	359.2	74.3	4.6	8.5	83.6	77.3	249.7	
LR18	syenite	47.92	108.56	10/10	400	324.3	59.4	5.4	9.4	66.4	63.5	157.9	
LR19	syenite	47.92	108.56	8/8	300	325.7	61.3	5.5	8.4	80.6	65.5	161.4	
LR21	syenite	47.92	108.55	7/8	400	339.1	58.3	5.4	7.7	94.4	72.5	138.5	
LR22	syenite	47.90	108.58	8/8	900	32.5	58.7	12.1	18.7	16.6	65.2	349.4	
LR23	syenite	47.92	108.57	9/9	500	347.8	59.4	4.7	7.8	96.4	78.3	124.4	
LR24	syenite	47.92	108.57	9/9	300	357.6	66.3	1.9	3.2	570.8	88.2	191.2	
LR25	syenite	47.93	108.58	9/9	500	348.2	71.9	5.9	9.7	62.2	78.6	217.2	
LR26	syenite	47.91	108.62	10/10	400	4.9	74.0	6.8	11.9	42.0	77.4	262.5	61.4
				Bearpaw Mountains: Central Intrusive Core									
BP4N	latite	48.23	109.66	12/15	200	354.5	69.9	3.3	5.9	175.0	83.4	220.7	
BP4R	latite	48.23	109.66	25/28	200	177.1	-62.2	3.6	9.8	65.7	84.8	94.2	
BP6	latite	48.18	109.64	10/10	150	351.8	77.5	5.8	8.6	78.8	71.5	239.8	

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BP7	shonkinite	48.21	109.65	13/13	100	350.8	76.9	2.9	5.4	209.0	72.4	236.5	
BP9	monzonite	48.20	109.65	6/7	100	342.8	73.6	2.8	3.1	588.8	75.0	215.1	49.9
BP10*	tinguaite	48.19	109.65	5/5	150	246.4	39.7	20.7	29.7	6.5	--	--	
BP12	shonkinite	48.17	109.69	9/9	100	335.3	62.7	3.9	5.8	172.1	72.5	156.1	
BP13	pyroxenite	48.17	109.70	12/12	100	178.9	-66.1	3.7	6.6	140.1	89.2	181.9	53.4
BP14	syenite	48.17	109.71	8/9	100	163.3	-78.8	3.2	4.9	235.5	68.1	233.7	51.6
BP23	shonkinite	48.24	109.52	7/8	200	349.6	68.6	6.6	8.2	83.8	82.4	193.1	
BP24*	latite	48.22	109.51	5/10	400	9.3	74.3	15.9	16.8	24.1	--	--	
BP34	shonkinite	48.17	109.76	10/10	100	8.1	66.0	2.6	4.5	296.9	84.6	336.0	
BP36	shonkinite	48.22	109.25	9/10	300	199.1	-67.8	3.8	5.8	173.6	77.3	322.0	
BP37	shonkinite	48.31	109.24	9/9	200	163.6	-76.9	5.3	8.7	76.9	71.1	229.2	
BP38	latite	48.21	109.10	8/8	100	147.2	-67.7	2.8	4.3	306.7	68.7	180.0	
BP39*	latite	48.21	109.10	8/8	400	211.3	46.0	10.0	15.4	24.5	- 9.4	223.0	
BP40*	shonkinite	48.21	109.10	7/7	100	132.6	74.5	15.1	21.5	12.4	25.7	85.7	
BP41	latite	48.26	109.35	10/10	200	160.6	-70.5	12.4	19.2	16.1	76.4	196.1	
BP42	latite	48.29	109.36	7/9	100	159.4	-60.9	2.7	4.2	327.8	74.2	144.7	
BP43*	latite	48.30	109.27	7/8	300	195.0	-34.8	15.2	23.4	10.6	--	--	
BP44	latite	48.23	109.11	9/9	100	351.8	74.6	3.0	5.0	238.3	76.3	234.0	
BP45	latite	48.23	109.66	6/9	100	335.4	66.1	9.1	11.8	39.5	73.7	170.5	
BP46	shonkinite	48.28	109.63	7/7	600	152.6	-44.1	3.8	5.4	195.0	59.0	123.9	
				Bearpaw Mountains: Northern Volcanic Field									
BP1	mafic phonolite	48.35	109.67	14/15	300	143.6	-71.4	3.9	8.2	91.4	66.7	193.4	
BP2	mafic phonolite	48.38	109.68	10/11	200	186.7	-43.7	4.0	6.4	146.4	66.6	54.9	
BP3	mafic phonolite	48.24	109.67	10/11	100	71.1	77.5	5.0	8.7	78.6	50.3	287.3	
BP5	mafic phonolite	48.24	109.67	10/12	100	10.5	36.1	3.8	7.3	112.8	60.6	49.9	
BP8*	mafic phonolite	48.25	109.67	7/8	100	129.6	80.9	11.6	17.9	18.1	35.5	267.1	
BP15	latite	48.28	109.59	8/9	100	3.3	45.7	2.7	4.1	335.8	68.7	62.3	
BP16*	mafic phonolite	48.28	109.56	7/7	200	57.4	78.8	29.3	42.5	3.3	--	--	
BP17	latite	48.28	109.57	8/8	100	308.4	79.5	5.3	7.6	98.4	57.5	220.0	
BP18	mafic phonolite	48.26	109.79	6/8	200	147.2	-50.7	11.4	14.9	25.0	60.0	138.1	
BP19*	mafic phonolite	48.28	109.81	9/9	300	209.8	-23.4	9.8	16.0	22.9	46.2	25.6	
BP20	mafic phonolite	48.36	109.67	9/9	400	168.6	-54.4	8.4	14.6	27.8	74.1	106.7	
BP21	mafic phonolite	48.51	109.64	11/11	100	151.5	-67.2	2.4	4.5	298.9	71.4	175.8	
				Bearpaw Mountains: Southern Volcanic Field									
BP11	mafic phonolite	48.18	109.64	10/10	100	25.6	78.6	6.2	10.3	51.3	66.3	274.0	
BP25*	mafic phonolite	48.18	109.49	9/9	100	34.3	38.8	3.5	5.8	175.4	52.1	12.2	
BP26*	mafic phonolite	48.17	109.50	6/7	100	350.0	33.2	17.1	22.3	11.1	--	--	
BP27*	mafic phonolite	48.13	109.57	8/9	600	32.9	-50.6	5.4	8.2	85.2	- 5.2	222.6	
BP28*	mafic phonolite	48.17	109.66	6/9	1000	102.8	-48.2	5.8	8.2	82.8	29.5	172.4	
BP29*	mafic phonolite	48.11	109.64	7/7	400	302.9	35.9	25.3	39.6	3.8	--	--	
BP30	mafic phonolite	48.09	109.70	9/9	100	46.6	69.3	4.8	7.9	94.1	60.5	313.1	
BP31*	mafic phonolite	48.06	109.56	8/8	200	72.9	-50.9	3.2	5.0	231.8	12.8	193.8	
BP32	mafic phonolite	48.04	109.55	8/9	400	132.7	-69.1	9.7	13.8	29.9	60.0	187.2	
BP33	mafic phonolite	48.04	109.38	6/7	100	165.5	-70.6	5.7	7.4	100.6	78.7	205.2	
BP35*	mafic phonolite	48.18	109.31	9/9	400	347.7	-32.3	9.5	15.6	24.0	-23.4	263.5	
				Highwood Mountains									
HW1	shonkinite	47.48	110.61	8/9	200	13.2	55.6	3.6	5.5	188.8	75.0	23.8	
HW2	phonolite	47.50	110.51	8/8	100	336.5	37.8	2.5	3.8	389.5	57.6	113.4	
HW5*	syenite	47.48	110.58	8/8	200	118.21	50.4	2.8	4.0	353.1	6.2	298.8	
HW6*	monzonite	47.48	110.62	9/9	150	280.4	40.2	33.8	57.8	1.9	--	--	

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TABLE 2. (continued)

Site	Lithology	λ	ϕ	N/No	Ho	D	I	α_{95}	δ	k	PLAT	PLONG	Age, m.y.
					Highwood Mountains (continued)								
HW7	shonkinite	47.47	110.61	9/9	200	0.37	54.6	2.4	4.0	362.0	77.7	68.0	
HW8	phonolite	47.42	110.58	7/7	200	320.9	55.8	5.4	7.7	95.8	59.3	152.7	
HW9	phonolite	47.40	110.57	7/7	200	315.2	75.2	2.3	3.3	518.2	61.1	206.7	
HW10	phonolite	47.40	110.58	7/7	200	341.5	72.5	7.0	6.3	142.5	74.8	209.4	
HW11	phonolite	47.32	110.60	8/8	100	340.4	78.3	11.2	17.3	19.3	67.5	229.7	
HW12	phonolite	47.60	110.34	9/9	200	357.4	59.6	2.9	4.5	289.2	82.6	85.4	
HW13	shonkinite	47.51	110.37	8/8	200	348.2	73.6	2.5	3.8	402.5	76.1	224.0	
HW14*	phonolite	47.57	110.44	7/7	150	40.3	60.6	18.5	26.5	8.1	--	--	
HW15*	phonolite	47.57	110.44	6/6	200	286.0	58.5	31.5	41.7	3.3	--	--	
HW16	phonolite	47.56	110.46	4/7	200	350.8	47.8	6.4	6.4	121.5	70.0	93.7	
HW17	phonolite	47.56	110.46	6/6	150	48.2	74.7	8.4	9.2	64.3	59.7	294.7	
HW18	phonolite	47.48	110.50	7/7	200	15.1	44.9	7.2	8.9	71.4	65.8	34.7	
HW19	phonolite	47.56	110.55	6/6	100	356.1	67.4	7.5	8.2	81.4	86.2	208.4	
HW20*	phonolite	47.56	110.46	6/6	200	26.0	34.0	4.9	6.3	139.0	54.2	24.4	
HW21	phonolite	47.56	110.46	6/6	300	3.7	72.6	6.4	8.3	79.4	79.4	260.3	
HW22	phonolite	47.54	110.49	7/7	200	3.3	36.6	2.7	3.8	396.7	62.7	62.7	
HW23	phonolite	47.56	110.47	10/10	300	36.3	66.0	4.0	6.3	147.2	65.9	324.1	
HW24-25-42	phonolite	47.54	110.29	27/27	100/200	330.0	72.0	1.8	5.4	217.9	71.1	199.9	50.1
HW26-27	shonkinite	47.54	110.29	9/13	100	315.1	69.1	2.8	5.9	285.5	61.2	186.9	50.1
HW28	shonkinite	47.52	110.25	8/8	150	351.0	56.2	2.5	3.9	385.1	77.3	104.6	
HW29	shonkinite	47.52	110.25	8/8	100	346.1	69.2	2.4	3.7	411.8	79.7	195.9	
HW30	shonkinite	47.58	110.23	4/11	200	333.0	70.3	5.7	5.7	149.8	71.8	191.9	50.4
HW31	phonolite	47.50	110.22	5/8	600	315.4	57.2	7.9	9.2	62.2	56.3	159.7	
HW32	phonolite	47.50	110.22	5/8	800	326.4	63.7	3.6	4.2	295.6	66.9	167.0	
HW33	shonkinite	47.46	110.63	8/8	100	8.2	40.1	2.5	3.8	391.0	64.5	51.7	
HW34	shonkinite	47.46	110.63	7/7	150	23.2	64.4	5.6	8.0	89.0	73.4	334.9	
HW35*	phonolite	47.49	110.63	7/7	200	33.1	51.0	32.9	48.1	2.6	--	--	
HW36	phonolite	47.39	110.42	8/8	200	312.9	59.0	4.9	7.5	102.2	55.6	164.5	
HW37*	phonolite	47.41	110.41	6/6	200	68.6	17.5	42.0	56.7	1.8	--	--	
HW38	phonolite	47.46	110.43	6/10	200	339.2	65.9	6.7	8.7	72.7	76.0	170.1	53.1
HW39	phonolite	47.46	110.44	8/8	400	308.4	59.5	5.7	8.8	74.9	52.9	168.0	
HW40	shonkinite	47.61	110.29	8/8	200	353.3	63.6	2.2	3.3	528.9	84.8	134.1	
HW41	shonkinite	47.64	110.50	9/10	200	354.6	71.0	2.6	4.3	314.0	81.4	228.5	
					Eagle Buttes								
EB1	latite	47.84	110.01	11/11	200	335.1	63.0	4.3	7.8	98.2	72.5	158.3	50.3
EB2	shonkinite	47.84	110.01	8/8	500	323.0	59.7	4.0	6.1	153.6	62.8	158.3	
EB3	shonkinite	47.87	110.03	7/8	200	356.7	65.6	3.4	4.8	240.8	87.8	158.2	51.6
EB4	shonkinite	47.87	110.03	5/6	200	329.1	62.6	9.1	10.5	74.8	68.1	159.9	
EB5	shonkinite	47.87	110.03	7/8	400	330.9	66.6	7.6	11.6	42.9	70.8	174.4	
EB6	shonkinite	47.84	110.01	7/8	400	17.0	68.8	6.9	9.9	57.5	78.3	312.0	
EB7	shonkinite	47.82	110.05	8/8	150	332.0	74.0	7.9	12.1	39.4	69.7	207.9	
EB8*	shonkinite	47.82	110.05	9/9	200	177.4	81.8	11.6	19.04	16.2	31.7	250.8	

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Sweetgrass Hills													
SH1	syenite	48.84	111.12	7/8	300	143.0	-73.8	6.2	7.6	96.7	66.3	200.3	
SH2-4	minette	48.84	111.12	11/13	300	176.0	-62.3	6.4	7.1	131.6	84.0	98.0	
SH3	minette	48.84	111.12	8/8	300	173.4	-59.5	4.2	5.8	173.4	80.3	100.4	51.2
SH5*	monzonite	48.84	111.12	9/9	300	242.6	-44.9	9.5	13.9	30.4	8.5	325.5	53.6
SH6	monzonite	48.87	111.43	6/8	600	128.1	-57.4	6.8	7.5	97.2	51.6	162.5	
SH7	minette	48.87	111.41	6/8	800	161.8	-44.1	12.1	15.8	22.1	63.0	106.8	
SH8*	minette	48.89	111.40	10/10	150	267.9	57.6	25.2	36.4	4.6	--	--	
SH9*	monzonite	48.87	111.40	8/8	200	160.8	-41.9	84.9	68.3	1.4	--	--	
SH10	minette	48.91	111.43	6/8	300	136.1	-60.0	5.8	5.8	157.5	58.4	160.8	
SH11	minette	48.88	111.42	6/7	300	13.1	66.2	8.8	9.6	58.8	81.4	335.8	
SH12	trachyte	48.84	111.16	8/8	600	163.0	-33.7	13.4	17.9	17.7	56.7	99.2	
SH13	syenite	48.84	111.12	7/8	150	152.3	-57.1	14.7	17.8	17.8	67.1	140.1	
SH14	syenite	48.84	111.12	5/6	150	168.1	-62.4	9.2	8.6	70.8	80.3	131.5	52.5
Missouri Breaks													
MB-1	peridotite	47.63	110.22	9/9	100	354.2	71.6	4.1	6.3	145.1	80.6	229.8	48.3
MB-2	peridotite	47.84	109.18	5/10	200	334.5	70.6	4.2	4.9	221.1	78.1	202.9	48.4
MB-3	peridotite	47.83	109.22	7/7	150	2.1	73.1	2.4	3.5	470.1	79.0	114.9	
MB-4	peridotite	47.86	109.00	11/11	200	0.6	73.4	5.2	9.6	64.9	79.0	110.5	
MB-5*	peridotite	47.88	108.99	4/8	300	110.6	-62.3	5.4	5.4	168.8	43.0	183.0	49.7
MB-6*	kimberlite	47.85	108.70	7/8	150	120.3	54.9	14.4	20.5	13.5	8.9	296.7	
MB-7	peridotite	47.92	108.52	6/12	150	31.0	64.6	9.7	12.5	35.1	69.0	338.8	
MB-8*	alnoite	47.88	108.42	15/15	150	192.0	-62.5	29.5	67.9	1.5	--	--	
MB-9	lamprophyre	47.02	108.60	8/8	100	10.2	74.8	5.0	7.7	97.5	74.4	126.8	
MB-10	alnoite	47.00	108.59	9/12	250	163.8	-67.2	3.7	6.0	163.1	78.9	182.8	50.2
Little Belt Mountains													
LB1	shonkinite	47.23	110.74	8/8	100	182.7	-72.6	2.5	3.8	399.4	79.2	257.0	51.1
LB2	minette	46.79	110.68	8/9	200	173.4	-62.5	2.8	4.2	319.7	84.5	129.0	54.1
LB3	quartz latite	46.87	110.67	10/10	200	13.4	67.1	3.1	5.4	204.3	80.6	316.1	55.3
LB5	lamprophyre	46.83	110.69	8/10	500	172.3	-48.5	4.4	6.7	128.9	71.6	91.0	
LB6	minette	47.06	110.72	8/8	200	7.9	56.3	3.7	5.7	178.8	78.3	36.7	59.2
LB7*	minette	46.85	110.69	8/9	200	226.0	-57.6	7.1	10.9	48.5	55.6	336.8	
LB8	syenite	47.22	110.73	9/11	400	207.3	-62.2	4.0	6.5	137.9	70.6	340.2	
LB9*	lamprophyre	46.80	110.70	8/8	400	32.6	65.4	27.2	42.7	3.3	--	--	
LB10	lamprophyre	46.81	110.70	8/8	300	166.5	-70.4	7.9	12.1	39.2	78.5	206.4	
LB11	lamprophyre	46.81	110.70	8/9	300	17.4	67.3	4.1	5.8	166.6	78.0	316.7	
LB12	rhyolite	46.97	110.72	6/7	900	310.5	62.5	6.3	8.2	81.9	55.7	172.7	50.9
LB13	rhyolite	46.96	110.72	7/8	300	1.2	79.2	9.3	11.4	43.1	67.9	250.4	50.9
LB14	quartz latite	47.05	110.58	8/8	200	341.2	66.2	2.8	4.3	307.6	77.3	173.5	54.2
LB15*	granite	47.06	110.59	8/8	200	74.5	51.8	7.0	10.0	56.4	33.1	325.6	47.9

λ, latitude (°N) of sampling site; ϕ, longitude (°W) of sampling site; N/No, ratio of number of samples collected to those used in determining the site mean direction; Ho, optimum demagnetizing field in oersteds; D and I, mean declination and inclination for each site; α_{95} and K, Fisher [1953] statistical parameters; δ, angular standard deviation; PLAT and PLONG, latitude (north positive) and longitude (east positive) of VGP.

*Sites not included in calculation of mean pole.

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to correct for possible declination errors resulting from local magnetic effects.

Remanent magnetization was measured on a Schonstedt SSM-1A spinner magnetometer. All samples were magnetically cleaned using the alternating field (af) method. Orthogonal vector demagnetization diagrams typical of results from the various lithologies throughout the north-central Montana province are shown in Figure 2. Overall, magnetic stability was good, and in most instances, it was possible to isolate a single component of magnetization (a component whose vector endpoint decays linearly to the origin with increasing peak demagnetization) despite varying amounts of secondary overprints. The secondary component of magnetization most often removed during demagnetization had a direction that pointed to the north and down, suggesting a present-day viscous overprint; no other consistent secondary overprints were observed.

The initial af demagnetization level used for each site was based on the minimum angular divergence between two or more pilot specimens and on their directional stability using vector end point plots. All samples were then demagnetized to one, two, and, for some, three higher peak fields. The peak field which maximized k (estimate of the precision parameter of Fisher [1953]) was selected as the optimum demagnetization level for that site; demagnetization levels between 20 and 40 mT were most common.

A summary of the pertinent paleomagnetic data after magnetic cleaning for the major igneous centers is given in Table 2. Many of these data have been reported by Diehl et al. [1980] and Jacobson et al. [1980]. For the Highwood Mountains, five sites from Shonkin Sag laccolith were combined into two composites sites, and in the Sweetgrass Hills, two sites located in exposures of the same lithology were combined. This reduced N to 169. Site-mean directions for the northern and southern volcanic fields of the Bearpaw Mountains have been corrected for geological dip. Data from three sites not reported by Jacobson et al. [1980] for the Judith Mountains also have been included in Table 2. These sites were located in an intrusion initially thought to be 78 m.y. old, but more recent dates indicate a 67 m.y. age, so these sites have been included for final analysis.

Eighteen sites for which the circle of 95% confidence exceeded our arbitrary cutoff value of 15° were rejected for final statistical analysis; 13 of these had very high scatter (α_{95}>25°). These sites either had abnormally high intensities and random directions, suggesting remagnetization by lightning, or were located in weathered (altered) lithologies and showed unstable behavior upon demagnetization. One additional site in the Little Rocky Mountains had remanent intensities too weak to be measured. To further minimize dispersion not related to the stable ancient geomagnetic field, sites whose virtual geomagnetic pole (VGP) was more than 38° removed from their respective group means (i.e., Eocene or Paleocene) also were rejected. The angle 38° was selected because it represents approximately twice the greatest angular dispersion, S_u, shown to be inherent in paleosecular variation over the past 5 m.y. for the paleolatitude of the sampling area [McElhinny and Merrill, 1975]. Twenty sites were eliminated by this test. Many of these sites had very divergent, intermediate directions, and we surmise that they acquired their magnetization during a polarity change or excursion of the geomagnetic field. Some may also have been tectonically rotated after magnetization. The latter is the most probable explanation for some sites located in the northern and southern Bearpaw volcanic fields where rotations about vertical axes may have occurred during gravity sliding. Last, individual specimens (116) that diverged from the site mean by an angle exceeding twice the angular standard deviation of the site also were rejected. These probably represent lightning strikes, rotated blocks, or gross orientation error.

Average directions of remanent magnetization and mean pole positions have been calculated for each igneous complex (Table 3). The mean pole reported for each complex is the mean of the site VGP's. The mean poles reported for the Bearpaw, Highwoods, and Little Rocky mountains in this study replace those reported by Diehl et al. [1980] and Jacobson et al. [1980]. We also have calculated a mean Eocene paleomagnetic pole and a mean Paleocene paleomagnetic pole from the north-central Montana data (Table 3), each being the vector mean of the corresponding site poles. In calculating a mean Paleocene pole, we have included one site (LB6) from the Little Belt Mountains that has a radiometric date indicating a Paleocene age. (This site was not used in calculating the mean direction and pole for the Little Belt Mountains.) Therefore, N is equal to 36 instread of 35 total if sites from only the Moccasin, Judith, and Little Rocky mountains had been used in the Paleocene pole calculations. The Paleocene pole calculated herein differs by less than 1.5° from that originally reported by Jacobson et al. [1980].

Our sampling scheme was such that each site probably represents an independent spot reading of the early Tertiary geomagnetic field. The apparent age span sampled, the between-site dispersion, and the presence of both normal and reverse polarities indicate that secular variation has been adequately sampled and averaged into our Eocene and Paleocene poles. Figures 3 and 4 show the distribution of Eocene and Paleocene VGP's and the mean paleomagnetic pole for each.

Last, since these poles differ by only 1.6° of arc and are not significantly different statistically, we have calculated a mean early Tertiary paleomagnetic pole for cratonic North America giving unit weight to the 130 site poles (Table 3).

Tilt Corrections

A problem of some concern when working with intrusive rocks is the possible presence of post-magnetization tilt. For the north-central Montana intrusions we believe this problem to be minimal. Regional tilt associated with late Laramide deformation can be ruled out since Upper Cretaceous-lower Tertiary sedimentary rocks in the plains surrounding the igneous centers are very nearly horizontal; for instance, Reeves [1928] reports a regional tilt to the northeast of less than a degree between the Highwood and Bearpaw mountains. In the Highwood Mountains the

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TABLE 3. Mean Directions of Remanent Magnetization and Paleomagnetic Pole Positions for Eocene and Paleocene Igneous Centers, North-Central Montana

	N	Mean Direction D	I	k	α_{95}	δ	Mean Poles Latitude	Longitude	K	A_{95}	δ
Bearpaw Mountains	30	349.2	66.9	33.7	4.6	13.8	82.0°N,	188.9°E	18.9	6.2	18.4
Highwood Mountains	30	348.8	63.8	26.6	5.2	16.6	82.1°N,	163.0°E	14.9	7.0	20.7
Sweetgrass Hills	10	338.8	58.8	31.7	8.7	13.7	73.7°N,	138.4°E	19.4	11.2	17.5
Eagle Buttes	7	339.2	66.7	82.9	6.7	8.2	76.5°N,	174.9°E	33.5	10.6	13.0
Little Belt Mountains	10	356.0	67.3	45.7	7.2	11.4	85.3°N,	219.8°E	21.2	10.7	16.8
Missouri Breaks	7	359.9	71.6	121.8	5.5	6.8	85.7°N,	157.7°E	34.4	10.4	12.8
Little Rocky Mountains	21	346.4	63.3	41.7	5.0	12.3	81.1°N,	153.0°E	21.3	7.0	17.2
Moccasin-Judith Mountains	14	347.2	71.0	58.2	5.2	10.2	77.8°N,	215.6°E	22.2	8.6	16.6
Eocene sites	94						82.0°N,	170.2°E	18.6	3.5	18.8
Paleocene sites	36						81.8°N,	181.4°E	20.2	5.4	17.8
All sites	130						82.0°N,	173.3°E	19.1	2.9	18.5

N, number of sites; D and I, mean declination and inclination; α_{95} and k, statistical parameters [Fisher, 1953]; δ, angular dispersion [Wilson, 1959].

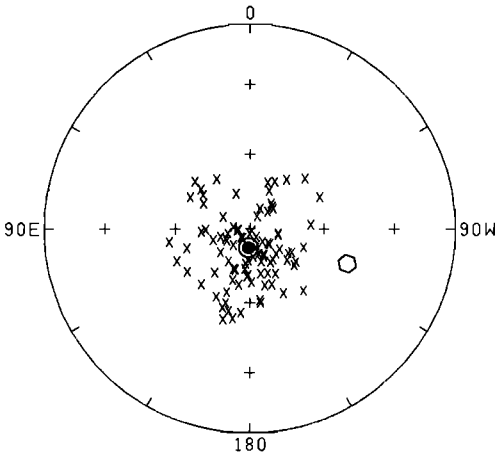


Fig. 3. Polar equal-area projection showing Eocene VGP's for the north-central Montana alkalic province and the mean paleomagnetic pole (solid circle) with its circle of 95% confidence. Mean pole is located at 82.0°N, 170.2°E (A_{95} = 3.5°). Hexagon is sampling location.

attitude of sediments near the sampling sites is generally horizontal, and in all cases, observed dips are less than 5° and random in direction. Tilt, where present, was probably the result of intrusion [Hurlbut and Griggs, 1939]. Therefore postmagnetization tilt appears not to be a problem with these intrusions. For isolated laccolithic intrusions, such as the Sweetgrass Hills, symmetry of sediment dips on the flanks of these intrusions indicate that they are substantially undisturbed. Tilting also appears to be insignificant in the Eagle Buttes and Missouri Breaks regions. In both regions, intrusions postdate the faulted folds that are genetically related to gravity sliding off the Bearpaw Mountains arch, as shown by dikes and plugs which cross the faults and sills which crop out on the uptilted side of a faulted fold but do not appear at equivalent stratigraphic horizons on the other side of the fault. Control on postintrusion tilting in the Bearpaw, Moccasin, Judith and Little Rocky mountains is less certain. For many sites, tilting can be shown to be minimal, but for other sites we are less sure. Since the angular dispersion of site-mean directions from each of these areas is less than that of the Highwood Mountains (for which undetected tilts are not a problem), large uncorrected tilts, which would greatly increase the dispersion, are unlikely. Asymmetric tilting of the Bearpaw intrusive core can also be ruled out since the dip of the sediments along the flanks of the core shows no asymmetry. In the Little Belt Mountains many of the earlier intrusions have been tilted as a large central igneous pluton was emplaced during the waning phases of activity [Witkind, 1973]. The degree of tilting is unknown. To minimize effects of tilting, sampling for the most part was restricted to the flanks of the Little Belt Mountains. Most measured tilts are less than 5°; the greatest is 9°. Correcting for dip did not significantly reduce the scatter of mean direction. In fact, the precision parameter of Fisher [1953] actually decreased from 36.0 to 30.4 with tilt correction, an amount that is not statisti-

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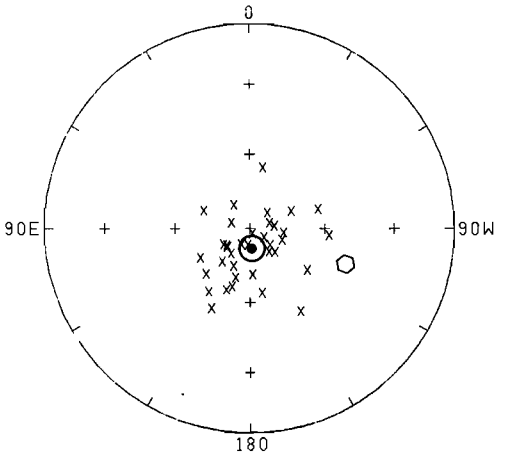


Fig. 4. Polar equal-area projection showing the Paleocene VGP's for the north-central Montana alkalic province and the mean paleomagnetic pole (solid circle) with its circle of 95% confidence. Mean pole is located at 81.8°N, 181.4°E (A_{95} = 5.4°). Hexagon is sampling location.

cally meaningful at the 95% confidence level [Cox, 1969], while the mean direction for the complex changed by less than 2.5°. Hence tilt is surely present, but since the change in direction is not significant and since no marked improvement in scatter occurred with correction, the magnitude is not sufficient to bias the data.

Time and Duration of Magnetization

None of the rocks drilled in the Highwood Mountains, Eagle Buttes, and Little Rocky Mountains are reversely magnetized, whereas both normal and reversely magnetized rocks were sampled at the other igneous complexes. Recent geomagnetic polarity time scales using the new K-Ar constants [Harrison et al., 1979; Mankinen and Dalrymple, 1979; Ness et al., 1980; Butler and Coney, 1981] show at most two or three normal polarity intervals (anomalies 22, 23, 24) for the period of igneous activity that built the Highwoods (50-53 m.y. B.P.) and Eagle Buttes (50-52 m.y. B.P.) and three normal polarity intervals (anomalies 27, 28, 29) for the period of intrusion (60-67 m.y. B.P.) responsible for the Little Rockies. Thus we do not know if these complexes became magnetized during one or several normal polarity intervals of the earth's magnetic field. However, several observations can be made.

1. For the Highwood Mountains most of the intrusives sampled belong to the second episode of magmatic activity and give radiometric ages that cluster around 50-51 m.y. Thus these intrusives probably were emplaced during a single normal interval. If this is the case, then magmatic activity associated with both the Highwood and Eagle buttes lasted no longer than 1.7 m.y. and possibly much less, judging from the above polarity time scales.

2. The most reliable K-Ar and fission track ages from the Little Rocky Mountains suggest emplacement between 63.9 and 61.0 m.y. [Marvin et al., 1980], a 2.9-m.y.-age range, which is mainly a reversed-polarity interval containing a short normal interval of 0.7 m.y. maximum duration. A span of less than a million years for the emplacement of these rocks appears unlikely which suggest that the intrusions probably occurred over several normal polarity intervals during the early Paleocene. However, some of the radiometric ages may have been slightly lowered to the 63.9-61.0 m.y. range by hydrothermal alteration [Marvin et al., 1980]. The Little Rockies may actually have been emplaced during an earlier portion of the Paleocene, where normal polarity intervals tend to be of longer duration (∿1.2 m.y.). Therefore intrusion during one normal polarity interval also cannot be ruled out.

At several of the other complexes, one polarity tends to dominate, in contrast to the substantially equal numbers of reversed and normal sites that would be expected for the age span of each complex in reference to the polarity time scale. Sampling bias can be ruled out since most of the lithologies present in each complex were drilled. Therefore we suspect that the igneous pulses that built the north-central Montana province were probably of a duration comparable to or even shorter than the average length of a polarity interval for the early Tertiary, circa 10^6 years.

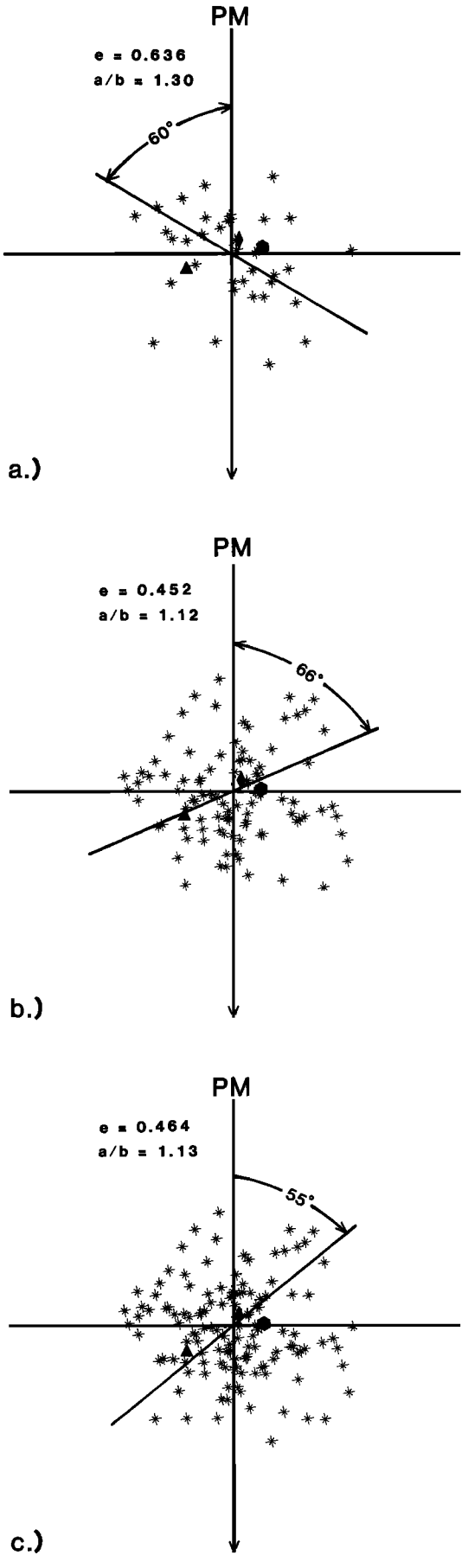
Discussion

The small angular difference (∿1.6° of arc) between the mean Eocene and Paleocene paleomagnetic poles calculated from the north-central Montana rocks suggests that the early Tertiary was a period of little or no discernible APW. To determine the extent of APW during the period of magnetization of these various igneous centers, we have undertaken a shape analysis of the north-central Montana VGP data. If APW was occurring during the magnetization process, the VGP data should be elongate along the expected direction of APW, for example, either along a great circle connecting the Cretaceous and the early Tertiary mean pole positions or between the early Tertiary position and the rotational axis. However, elongation may also be the result of the direction-to-pole mapping relationship, that is, a circular distribution of directions maps into an oval distribution of VGP's that is elongate normal to the paleomeridian.

Figure 5 is a plot of the Eocene, the Paleocene, and the combined Eocene-Paleocene VGP's for the north-central Montana province; the mean of each distribution has been rotated to the center of the projection. Visually the distributions appear to be circular; X^2 calculations (Table 4) indicate that for all three distributions, the VGP's are Fisherian at the 95% probability level.

However, objection has been raised as to the validity of the X^2 goodness-of-fit test [Baag and Helsley, 1974], so we have also analyzed the data using the technique of Engebretson and Beck [1978]. Engebretson and Beck have shown that information concerning the shape of any directional data set can be obtained by calculating the moments of inertia of the data set in question. From the moments of inertia, the eccentricity (e) and the ratio of the major to minor axis (a/b) of an ellipse that best approximates the distribution can be obtained. A distribution which is purely circular would have an eccentricity of zero (e = 0) and a major to minor axis ratio of

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one (a/b = 1). If the data lie on a line, an example of extreme elongation "e" would equal one and the "a/b" would approach infinity. Real paleomagnetic distributions should have eccentricity parameters that fall in between the extremes given above. For example, the distribution of VGP's from the Hawaiian data set of Doell and Cox [1965], which has been shown by Baag and Helsley [1974] to have a high degree of circularity, has eccentricity parameters of "e" = 0.295 and "a/b" = 1.05 [Engebretson and Beck, 1978].

The eccentricity parameters for the Montana data are shown in Figure 5. Elongation in these data sets obviously is very slight. As seen in Figure 5c, the combined Eocene/Paleocene data set is not quite as circular as the Hawaiian data set; this is also true for the Eocene and Paleocene data sets alone. What elongation that does exist in the Paleocene data (Figure 5a) is not along a great circle between the Cretaceous and Paleocene poles. In contrast, both the Eocene and combined Eocene/Paleocene data sets have slight elongations that are either along a great circle passing through the Cretaceous pole and/or the rotational axis or the Miocene/Oligocene pole (Table 6). Therefore elongation in these data sets might reflect some APW but also may be due in part to the direction-to-pole mapping function. The elongation is certainly minor, and since the Eocene and Paleocene poles are not statistically distinct and the combined data have an angular dispersion similar to that described by Fisher, we believe that the Montana data are consistent with an APW stillstand during the early Tertiary.

A compilation of paleomagnetic studies of Eocene and Paleocene rocks for stable North America is given in Table 5. Several studies have been omittted: (1) Strangway and McMahon's [1973] work on the weakly magnetized oil shales of the Green River Formation was not listed because these sediments yielded a very anomalous pole position, the probable result of authigenic pyrrhotite being the dominant magnetic mineral in these sediments [Sheriff and Shive, 1982], (2) the Beaverhead Valley volcanics [Hanna, 1967] and the Flagstaff Mountain intrusive and associated baked zone [McMahon and Strangway, 1968a, b] studies were eliminated due to uncertainties in age, (3) the work of Butler and Taylor [1978] on the Paleocene Nacimiento Formation was excluded because Barnes and Butler [1980] and Vugteveen et al. [1981] believe that these sediments are biased by an inclination error, and (4) last, we have chosen not to include the work of Currie and Larochelle [1969] on the Mistastin Lake volcanics. The $^{40}Ar/^{39}Ar$ age determinations on melt rocks associated with

Fig. 5. Centered equal-area projection of (a) Paleocene, (b) Eocene, and (c) Paleocene/Eocene VGP's for the north-central Montana alkalic province. PM is paleomeridian; "a/b" is major to minor axis ratio; "e" is eccentricity; hexagon is rotational axis, the triangle is mean Cretaceous pole (Table 6), and diamond is the mean Miocene/Oligocene pole (Table 6); additional axes indicate orientation of major axis.

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TABLE 4. χ^2 Calculation for Goodness of Fit of VGP's to a Fisher Distribution

	Paleocene		Eocene		Paleocene/Eocene	
	OBS	EXP	OBS	EXP	OBS	EXP
ϕ'						
1-40	2	4.00	10	10.4	12	14.4
41-80	3		11		14	
81-120	7		5		11	
121-160	6		10		17	
161-200	2		13		15	
201-240	1		10		11	
241-280	4		14		17	
281-320	7		7		15	
321-360	4	4.00	14	10.4	18	14.4
	$\chi^2 = 10.00^1$		$\chi^2 = 7.11^1$		$\chi^2 = 3.89^1$	
θ'						
0-3	1	1.0	0	2.4	2	3.4
3-7	4	4.0	14	9.8	16	13.9
7-11	7	6.1	17	15.0	28	21.2
11-15	8	6.8	14	16.9	19	23.7
15-19	3	6.1	12	15.8	16	21.9
19-23	5	4.8	11	12.7	15	17.4
23-27	3	3.2	11	9.1	14	12.3
27-31	2	2.0	10	5.8	11	7.7
31-35	3	1.1	3	3.4	7	4.4
>35	0	0.9	2	3.3	2	4.1
	$\chi^2 = 6.3^2$		$\chi^2 = 10.06^2$		$\chi^2 = 10.19^2$	

ϕ', azimuthal angle of VGP after rotating mean of the distribution to the center of the projection; θ', angular deviation of the VGP from the mean; χ^2, the value calculated according to Irving [1964].

[1]Expected value = 12.6 for 95% probability.

[2]Expected value = 14.7 for 95% probability.

the Mistastin Lake impact crater [Mak et al., 1976] define an age plateau in the range of 34-41 m.y., while conventional K-Ar dates on the volcanics [Wanless et al., 1972] indicate a 37-38 m.y. age. Since the age of these volcanics cluster at the Oligocene/Eocene boundary and are at least 6-7 m.y. younger than rocks from the studies listed in Table 5, we believe that the paleomagnetic results from these volcanics are more closely aligned with mid-Tertiary (22-38 m.y.) data than with the early Tertiary (45-65 m.y.) data that we have listed in Table 5.

TABLE 5. Early Tertiary Paleomagnetic Studies, North America

	Pole(°N/°E)	N	Age Range**	Reference
	Paleocene Studies			
Golden basalts, Colo.	72/64	2	~60.2	Larson et al. [1969]
Cape Dyer basalts, Canada	83/305	5	~59.5	Deutsch et al. [1971]
Gringo Gulch volcanics, Ariz.	77/202	25	63-67	Barnes and Butler [1980]
Montana intrusives	82/181	36	60-69	this study
	Eocene Studies			
Absaroka volcanics, Wyo.	84/177	19	45-49	Shive and Pruss [1977]
Rattlesnake Hills volcanics, Wyo.	79/146	18	45-50	Sheriff and Shive [1980]
Monterey intrusives, Va.*	89/0	5	-	Lovlie and Opdyke [1974]
Monterey intrusives, Va.*	86/244	11	42-47	Ressetar and Martin [1980]
Montana intrusives	82/170	94	47-54	this study

N, number of sites used in determining pole position.
*Lovlie and Opdyke [1974] originally reported data from six sites. However, the mean direction from site 2 was based on only two samples. We have not included this site in our analysis; therefore N = 5.
**Ages corrected to new IUGS constants.

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TABLE 6. North American Reference Poles: Cretaceous-mid-Tertiary

	N	R	K	A_{95}	PLAT	PLONG
Oligocene-early Miocene (22-38 m.y.)	102	94.10214	12.8	4.1	83.2°N	148.0°E
Middle-early Eocene (44-54 m.y.)	147	137.94584	16.1	3.0	82.8°N	170.4°E
Paleocene (55-67 m.y.)	68	65.70143	29.2	3.2	81.5°N	192.6°E
Cretaceous* (75-120 m.y.)	8**	7.98848	607.8	2.2	68.0°N	186.0°E

*Data from Mankinen [1978].
**Unit weight given each to study.

The pole positions listed in Table 5 vary in the degree to which each has averaged out secular variation. In fact, in several of the studies listed, it was the opinion of the original investigators that secular variation had not been adequately sampled. Therefore it would be unwise to give unit weight to each study listed in Table 5 in calculating a mean paleomagnetic pole for the Eocene and Paleocene. However, even though the sampling scheme in many of the studies may not have averaged out secular variation, each study does represent a certain number of independent spot readings of the geomagnetic field. In the absence of multiple sampling of a single rapidly magnetized rock unit, the number of independent spot readings of the ancient geomagnetic field must be approximately equal to the number of sites. Therefore in computing a mean Eocene and Paleocene paleomagnetic pole for the stable interior of North America, we have given unit weight to each site mean listed or calculated from the studies given in Table 5. The mean Eocene (44-55 m.y.) pole so calculated lies at 82.8°N, 170.4°E (A_{95} = 3.0°), while the mean Paleocene pole lies at 81.5°N, 192.6°E (A_{95} = 3.2°). Table 6 summarizes the pertinent paleomagnetic data.

These two early Tertiary paleomagnetic poles are separated by 3.3° of arc, a somewhat greater separation than the Eocene and Paleocene poles calculated from the Montana data alone. This slightly larger separation may signify that the data used in calculating the Eocene and Paleocene poles listed in Table 6 more completely average persistent nondipole components of the field [McElhinny and Merrill, 1975; Coupland and Van der Voo, 1980]. As seen in Figure 6, the Eocene and Paleocene paleomagnetic poles are significantly different from the present rotational axis and are statistically similar at the 95% probability level. The conditional test of McFadden and Lowes [1981] also indicates that these two data sets share a common mean at the 95% confidence level. The eccentricity parameters of this early Tertiary data set (e = 0.391; a/b = 1.09) indicate that this directional data set is more circular than the Montana data and nearly as circular as the Hawaiian data. What minor elongation exists is once again toward the rotational axis. We believe that this larger data set, like the Montana data, is consistent with the hypothesis that the early Tertiary was a time of little or no discernible apparent motion (stillstand) of the pole relative to North America. It may be that the time window represented by this early Tertiary data is not large enough to detect the effects of the apparent polar movement.

Also given in Table 6 and shown in Figure 6 is a mid-Tertiary reference pole for North America calculated from rocks ranging in age from 22 to 38 m.y. The data for this reference pole comes from ash flow units and associated rocks in Utah and Nevada [Gromme et al., 1972; Gose, 1970; Best et al., 1973; Shuey et al., 1976; Nairn et al., 1975], the volcanics of San Juan Mountains [Diehl et al., 1974; Beck et al., 1977; Tanaka and Kono, 1973; Ellwood, 1982], the Spanish Peaks dikes of Colorado [Larson and Strangway, 1969], basaltic lavas from State Bridge, Colorado [York et al., 1971], and the volcanics of Mistastin Lake in Canada [Currie and Larochelle, 1969]. For the Nevada and Utah ignimbrite studies and the San Juan volcanics,

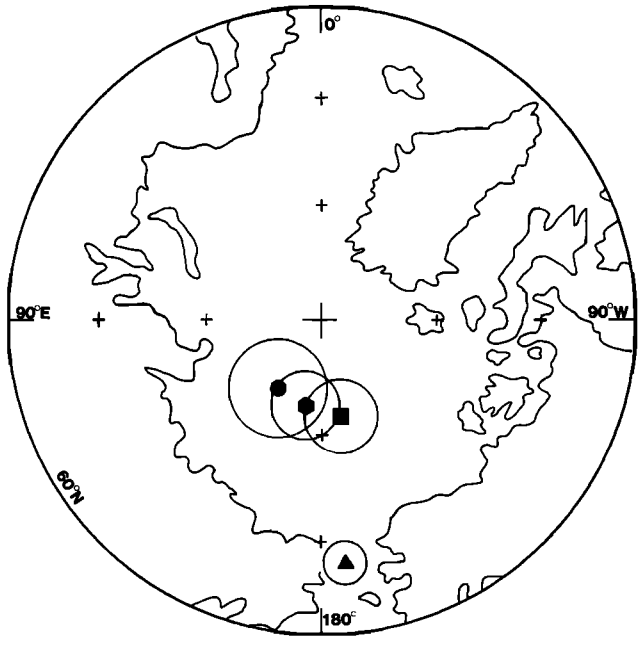


Fig. 6. Cretaceous (triangle), Paleocene (square), Eocene (hexagon) and Oligocene-early Miocene (circle) reference poles for stable North America including their circles of 95% confidence. Projection is equal-area.

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many of the volcanic units have been sampled by more than one investigator. (In utilizing the work of Nairn et al. [1975] only those results with circles of 95% confidence of 15° or less were used.) These multiple results for a single quick-cooling volcanic unit have been combined, so that no units except thick felsic flows, caldera fillings, and intrusive units are represented by more than a single entry. This procedure resulted in 102 site (unit) VGP's from which we have calculated an Oligocene-early Miocene reference pole for North America. The pole so calculated is located at 83.2°N, 148.0°E (A_{95} = 4.1°). This differs by some 6.2° of arc from the mid-Tertiary pole of Diehl et al. [1980] and is only 2.5° removed from the Eocene reference pole calculated above. In fact, this pole is not significantly different from the Paleocene pole at the 95% probability level. We believe this Oligocene-early Miocene pole is a better estimate of the mid-Tertiary paleomagnetic field that we previously reported [Diehl et al., 1980] in which unit weight was given to individual studies. One of those studies [Young and Brennan, 1974] was of rocks much younger (∿15 m.y.) than the other studies, and another study [Gilliland et al., 1969] reported only a pole position without the supporting data.

The newly calculated Oligocene-early Miocene pole indicates that no rapid shift of the pole occurred between 44 and 35 m.y. B.P., as previously suggested by Diehl et al. [1980]. Instead, it would appear that a gradual motion of the apparent pole was occurring during early to mid-Tertiary times and that this is only evident because the time window is sufficiently large.

As shown in Figure 6, the Paleocene and Eocene paleomagnetic poles are well removed from the Cretaceous pole (Table 6) for stable North America. Approximately 13° of arc separate the Paleocene and Cretaceous poles. Since the age of the youngest rocks yielding the Cretaceous pole position is ∿75 m.y. and the age of the oldest rocks yielding the Paleocene pole is ∿67 m.y., it would appear that the 13° of counterclockwise rotation required of North America occurred very rapidly, in about 10 m.y. The proximity of the Paleocene, Eocene, and mid-Tertiary poles and the absence of significant elongation in the early Tertiary data along a great circle between the Cretaceous and the early Tertiary positions of the pole indicate that this period of rapid apparent polar motion ceased by the end of the Cretaceous or earliest Paleocene (∿65 m.y.). Paleomagnetic results from the Roskruge volcanics [Vugteveen et al., 1981], dated at ∿72 m.y., are consistent with the above scenario, as the Roskruge pole (73.6°N, 176°E) lies midway between the Cretaceous and Paleocene poles. The onset of this rapid apparent polar motion may coincide with a major reorganization of plate motions at ∿80 m.y. [Coney, 1978] and with an increase in the absolute motion of the North American plate [Engebretson et al., 1982]. The initiation of this period of rapid apparent polar motion during latest Cretaceous correlates well with the beginning of the Laramide Orogeny, as noted by Vugteveen et al. [1981], which many investigators believe may be the expression of stresses imposed upon a rapidly moving North American plate between 80 and 40 m.y. B.P.

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